

Martins AWOJIDE

+250 (798) 694-256 | Kigali, RW (Relocating to Europe by H2 2026) | awojidemartins@gmail.com

PROFILE

Chemical Engineer (SPE scholarship winner) with MS in Engineering AI from Carnegie Mellon and Lean Six Sigma Black Belt certification. Combines process engineering fundamentals with operational excellence and AI/ML expertise to drive industrial transformation. Delivered **\$150K+** in verified savings through DMAIC projects while deploying SAP enterprise systems and edge ML solutions across multi-plant manufacturing operations. Proven adaptability through **3-year rotational assignments** across 3+ facility locations. Seeking to leverage chemical process optimization, continuous improvement rigor, and Industry 4.0 capabilities to accelerate TotalEnergies' renewable energy transition through OneTech's Graduate Program.

EDUCATION

Carnegie Mellon University

Pittsburgh, USA

Master of Science, Engineering Artificial Intelligence (with Distinction)

Aug 2023 — May 2025

- Cumulative GPA: 3.75/4.0 | Mastercard Foundation Scholarship
- **Focus Area:** AI-augmented process optimization — applying edge ML to industrial predictive maintenance, quality control, and real-time process analytics for manufacturing and energy systems
- Relevant Coursework: Efficient Machine Learning, Deep Learning, AI System Design, Embedded System Development, Systems Software Engineering, Cloud Computing

University of Lagos

Lagos, NGA

Bachelor of Science, Chemical Engineering (First Class Honours)

Oct 2011 — Dec 2016

- Cumulative GPA: 4.74/5.0 | Top 10% of Class
- **Society of Petroleum Engineers Star Scholarship** | MTN Foundation Scholarship | NNPC Agbami Professional Scholarship | National Merit Award
- Relevant Coursework: **Process Optimization, Process Dynamics & Control, Reaction Engineering**, Industrial Chemistry, Principles of Plant Design, Polymer Processing

TRAINING AND CERTIFICATIONS

Lean Six Sigma Black Belt, Benchmark X 360

Feb 2020

- **Batch Topper, LSSBB Mumbai '20 Cohort** (Top performer among 40+ candidates) 
- Certified in Lean, DMAIC, Statistical Process Control, Measurement System Analysis, and Change Management

Design for Six Sigma (DFSS) Training, Benchmark X 360

Jan 2021

- Advanced training in IDDOV (Identify, Define, Develop, Optimize, Verify) methodology for robust product and process design

AI Engineering Bootcamp, Andela

Feb 2026 — Apr 2026

- Production-grade LLM applications, RAG systems architecture, MLOps for industrial and operational use cases

PROFESSIONAL WORK EXPERIENCE

Industrial Process Optimization Engineer

Dec 2018 — Apr 2022

Dufil Prima Foods Plc (3-Year Rotational Assignment)

Lagos & Kaduna, NGA

Multi-site rotational role spanning Data Science/Industrial Optimization, Operations Management, and Business Intelligence across 3+ manufacturing facilities during initial orientation, followed by deep assignments in production optimization, warehouse/logistics operations, and strategic decision support

Process Engineering & Continuous Improvement (Dec 2018 — Dec 2020)

- Facilitated **DMAIC project** on noodle manufacturing process optimization, reducing fat content variability from **17.0%** to **16.5%** through **Measurement System Analysis (MSA)** and material balance optimization — direct application of chemical process fundamentals
- Led **DMAIC project** on laminate packaging process, reducing raw material wastage from **3.28%** to **1.3%** in six months, delivering **\$53,900** in Year 1 and **\$88,800** p.a. sustained savings through **statistical process control** and robust process design

- Architected **low-code paperless reporting system** deployed across 3 factory locations (Raffles, Insignia, Northern Noodles), reducing consumables cost by **95%** and enabling real-time process analytics — bridging Industry 4.0 digitization with lean principles
- Managed packaging reliability improvement project reducing product leakage from **3.2%** to **1.3%** through **root cause analysis** (Fishbone, 5 Whys, Pareto) and reliability engineering
- Conducted **kaizen events** and trained 40+ factory staff on 5S, Lean Six Sigma, and process fundamentals while mentoring Black Belt facilitators on DMAIC execution

Operations Management — Warehousing & Logistics (Jan 2021 — Nov 2021)

- Managed end-to-end warehouse and logistics operations for high-volume manufacturing facility, coordinating raw materials, packaging, finished goods, utilities, and spare parts across integrated supply chain
- Led **SAP EWM system design and deployment** for inbound/outbound logistics, collaborating with SAP functional consultants on warehouse structure configuration, bin management, and park management for 20+ core associates
- Developed **low-code data acquisition system** using Google AppSheet feeding SAP PM (Plant Maintenance) module for spare parts inventory and maintenance workflows, reducing manual data entry by **65%**
- Managed **120+ informal contract workers** in volatile economic conditions; implemented daily incentive structures maintaining 95% attendance during peak demand — workforce optimization under resource constraints
- Led teams through **ISO 22000 (FSMS)** and **ISO 9001:2015 (QMS)** audits achieving **zero** non-conformance, demonstrating quality systems excellence
- Reduced warehouse/logistics downtime from **0.85%** to **0.42%** YTD through **Theory of Constraints (TOC)** and **Value Stream Mapping (VSM)**
- Reduced truck turnaround time by **57%** through ETA tracking and **SMED** (Single-Minute Exchange of Die) in park management
- Deployed emergency **3x surge capacity** expansion in **72 hours** during supply chain crisis, preventing demurrage charges through rapid process redesign

Business Intelligence & Decision Support (Dec 2021 — Apr 2022)

- Supported Business Intelligence strategy for Head of Continuous Improvement & Sustainability at Group Corporate HQ
- Refined KPIs and OKRs, built data workflows automating appraisal systems for 14 operational units across 3 factory sites
- Analyzed financial and operational data using **statistical process control** to identify improvement opportunities for Six Sigma teams
- Facilitated cross-functional improvement projects using **root cause analysis** and hypothesis-driven problem solving

TECHNICAL SKILLS

Chemical & Process Engineering: Process optimization, reaction engineering, process dynamics & control, material & energy balances, statistical process control, process capability studies, measurement system analysis (MSA), design of experiments (DOE)

Operations Excellence & Lean: DMAIC, DFSS, IDDOV, 5S, Kaizen, Kanban, SMED, Value Stream Mapping (VSM), Theory of Constraints (TOC), Gemba walks, root cause analysis (Fishbone, 5 Whys, Pareto), quality systems management (ISO 9001, ISO 22000)

Industry 4.0 & Digital Transformation: Edge ML deployment, TinyML, Federated Learning, predictive maintenance systems, digital process monitoring, IoT sensor integration, MLOps for industrial applications, low-code/no-code development (AppSheet), SAP EWM deployment

Data Analytics & AI/ML: Python (pandas, scikit-learn, PyTorch), SQL, PySpark, statistical analysis, hypothesis testing, LLM/agent systems (LangChain, function calling), Minitab, Quality Companion, Excel (advanced)

Enterprise Systems: SAP EWM (warehouse management), SAP PM (plant maintenance), ERP system deployment and integration, KPI/OKR framework design

SELECTED PROJECTS & PUBLICATIONS

- *M. Awojide, E. Aiken. Inflation Nowcasting with Online Search Volume Data — Preprint* [↗](#)
- *G. Igwegbe, M. Awojide, A. Adetunji. Real-World Federated Learning on Edge Devices with NVFLARE — Deep Learning Indaba 2025 Poster* [↗](#)

- G. Igwegbe, M. Awojide, M. Bless. AutoVisionInspector — *Kaggle Competition* [↗](#)
- D. Nsanzimfura, C. Kwizera, G. Igwegbe, M. Awojide. SCADAgric: Agentic Farm Monitoring — *Repository* [↗](#)
- G. Igwegbe, M. Awojide, M. Bless, N. Kadzo. “Hello Afrika!”: Speech Commands in Kinyarwanda. Presented at Data Science Africa 2024 — *Conference Paper* [↗](#)
- G. Igwegbe, M. Awojide. TinyML Digital Counter for Electric Metering System. Hackster.io Impact Summit 2023 — *Article* [↗](#) *Community Spotlight* [↗](#)

ACADEMIC & RESEARCH EXPERIENCE

Graduate Teaching Assistant

Jan 2025 — Oct 2025

CMU-Africa

EAI System Design (Jan — Apr 2025): Advised 40+ graduate students on resource-constrained edge devices (Raspberry Pi, Arduino) for TinyML tasks and ModelOps exercises using AWS, Docker, Flask, and Plotly Dash

EAI Project Methods (Aug — Oct 2025): Guided students on AI engineering fundamentals including stakeholder analysis, system requirements, prototyping, Agile planning, and CI/CD methodologies

Product Research Assistant

May 2024 — Dec 2024

Embedded Systems Lab, CMU-Africa

Developed design strategy using IDDOV (DfX) framework for low-cost multimodal data acquisition device. Conducted Kano surveys in Africa and USA, built Kano model for MVP feature definition, and applied QFD to determine technical specifications for SBC/MCU selection.

LANGUAGES & ADDITIONAL INFORMATION

Languages: English (Native), French (Basic — actively improving for European collaboration)

Geographic Flexibility: UK-based with HPI visa pathway (relocating Aug-Sep 2026); prepared for France-based rotations and international mobility across TotalEnergies' European operations network

Community Leadership: Co-organizer of TinyML workshops across Nigeria, Rwanda, and Japan; active contributor to edge AI and continuous improvement communities in Africa